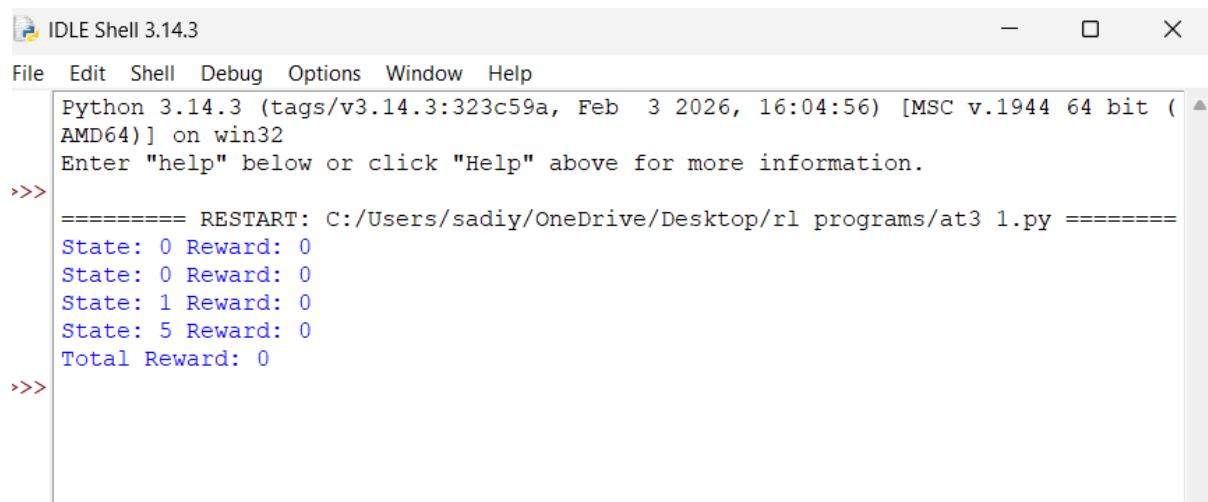


1. Grid World Navigation

```
import gym
import numpy as np
env = gym.make("FrozenLake-v1", is_slippery=False)
state = env.reset()[0]
done = False
total_reward = 0
while not done:
    action = env.action_space.sample() # Random action
    next_state, reward, done, truncated, info = env.step(action)
    print("State:", next_state, "Reward:", reward)
    total_reward += reward
    state = next_state
print("Total Reward:", total_reward)
```

Output

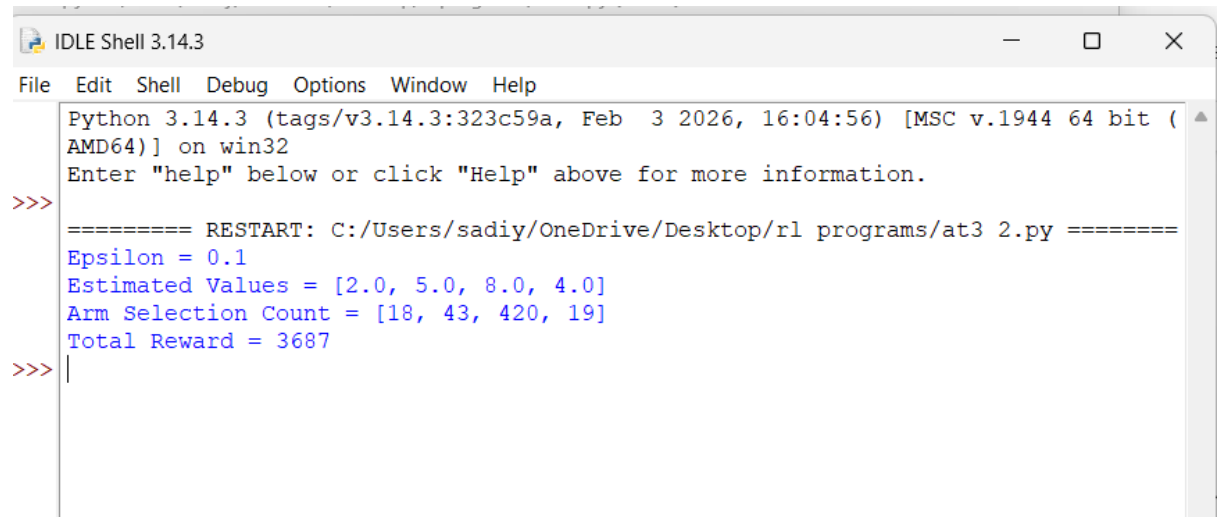


```
IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> ===== RESTART: C:/Users/sadiy/OneDrive/Desktop/rl programs/at3 1.py =====
State: 0 Reward: 0
State: 0 Reward: 0
State: 1 Reward: 0
State: 5 Reward: 0
Total Reward: 0
>>>
```

2. ϵ -Greedy Multi-Armed Bandit Algorithm (500 Trials)

```
import random
rewards = [2, 5, 8, 4]
epsilon = 0.1
counts = [0, 0, 0, 0]
values = [0, 0, 0, 0]
total_reward = 0
for i in range(500):
    if random.random() < epsilon:
        arm = random.randint(0, 3)
    else:
        arm = values.index(max(values))
    reward = rewards[arm]
    total_reward += reward
    counts[arm] += 1
    values[arm] = values[arm] + (reward - values[arm]) / counts[arm]
print("Epsilon =", epsilon)
print("Estimated Values =", values)
print("Arm Selection Count =", counts)
print("Total Reward =", total_reward)
```

Output



```
IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/sadiy/OneDrive/Desktop/rl programs/at3 2.py =====
Epsilon = 0.1
Estimated Values = [2.0, 5.0, 8.0, 4.0]
Arm Selection Count = [18, 43, 420, 19]
Total Reward = 3687
>>> |
```

3. Value Iteration Algorithm

```
import numpy as np

gamma = 0.9

V = np.zeros(4)

reward = [0, 0, 0, 10]

for i in range(10):

    newV = np.copy(V)

    for s in range(3):

        newV[s] = reward[s] + gamma * V[s + 1]

    newV[3] = reward[3]

    V = newV

print("Optimal State Values")

print(V)
```

Output:



The screenshot shows the IDLE Shell 3.14.3 window. The title bar is "IDLE Shell 3.14.3". The menu bar includes "File", "Edit", "Shell", "Debug", "Options", "Window", and "Help". The shell area displays the following text:

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/sadiy/OneDrive/Desktop/rl programs/at3 5.py =====
Optimal State Values
[ 7.29  8.1   9.  10. ]

>>> |
```

4. Smart Energy Management System

```
import random

rewards = [2, 5, 8, 4]

random_reward = 0

greedy_reward =

for i in range(100):

    arm = random.randint(0, 3)

    random_reward += rewards[arm]

best_arm = rewards.index(max(rewards))

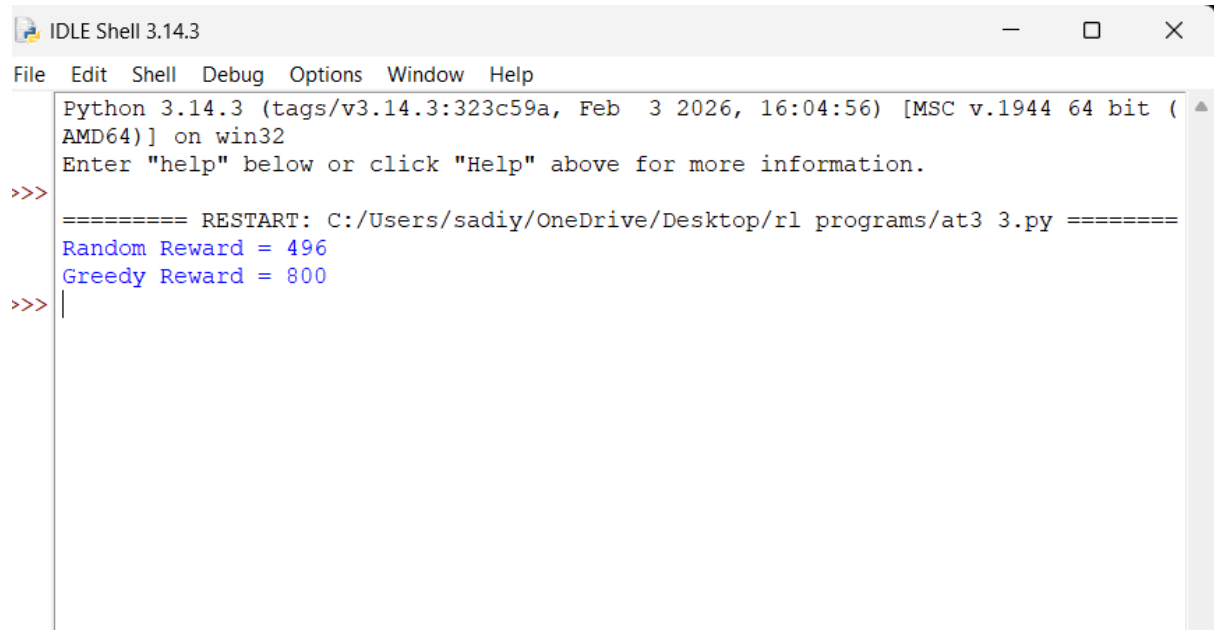
for i in range(100):

    greedy_reward += rewards[best_arm]

print("Random Reward =", random_reward)

print("Greedy Reward =", greedy_reward)
```

Output:



```
IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/sadiy/OneDrive/Desktop/rl programs/at3 3.py =====
Random Reward = 496
Greedy Reward = 800
>>> |
```

5. Smart Energy Management System

```
import random
threshold = 50
energy = 0
reward = 0
for i in range(20):
    use = random.randint(1, 5)
    if energy + use <= threshold:
        energy += use
        reward += 10
    else:
        reward -= 5
print("Energy Used =", energy)
print("Total Reward =", reward)
```

Output:

```
>>> Greedy Reward = 000
>>> ===== RESTART: C:/Users/sadiy/OneDrive/Desktop/rl programs/at3 4.py =====
>>> Energy Used = 49
>>> Total Reward = 155
>>>
```